RAG Advanced Topics

Learning Goals

By the end of this lesson, students should be able to:

- Explain why naive similarity search fails in RAG
- Distinguish symmetric vs asymmetric embedding use cases
- Understand why vector databases are needed

RAG Challenges

The core problem in RAG

- Most RAG systems assume: “If two texts are related, their embeddings should be similar.”
- This assumption is often false.
    - Queries and answers express different semantics
    - Similarity-only embeddings miss relevant documents
    - Retrieval errors propagate to generation
- Example:
    - Query: “Why is the sky blue?”
    - Passage: “Light scattering in the atmosphere causes blue wavelengths to dominate.”
    - These are related, but not semantically similar statements.
    - Pure similarity search penalizes this mismatch.

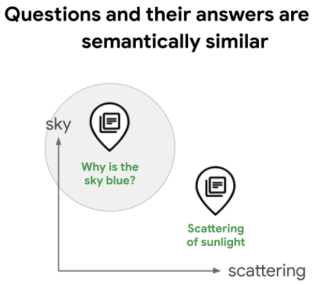


Figure 1. *Questions and their answers aren't semantically similar, which means that embeddings won't automatically show their relation.*

Source

The key insight

- Embeddings should be optimized for the task being performed.
- Not all embedding uses are the same. We might want to use them for:
    - Retrieval
    - Classification
    - Clustering
    - Similarity scoring
- Using the wrong embedding objective silently degrades RAG quality.

Symmetric vs asymmetric tasks

- Symmetric tasks
    - Text ↔ Text comparison
    - Same role on both sides
    - Example: similarity, clustering
- Asymmetric tasks
    - Query ↔ Corpus
    - Different roles and intent
    - Example: search, Q&A, fact checking
- RAG is fundamentally asymmetric.

How this problem is solved

- Task-aware embeddings: different projections are used for the two sides of the task
- This may involve separate models or a single model trained for asymmetric retrieval
    - Question embeddings align with answer embeddings
    - Claim embeddings align with evidence embeddings
    - Instruction embeddings align with implementation embeddings

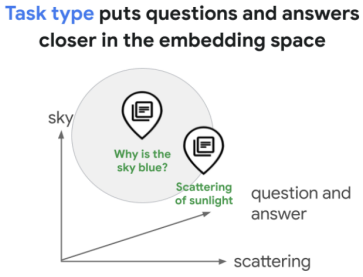


Figure 2. *Task types optimize embeddings for specific tasks. In this case, questions and answers are brought closer together in the embeddings space.*

Source

- Embedding symmetric content, context as metadata
    - Only embed the questions in a FAQ document and save the answers as metadata
    - Perform RAG using the same model for embedding the user questions
    - In he returned results, use the answers as the context
    - Works best when questions are stable and answers are long or variable
- Hypothetical Document Embeddings (HyDE)
    - Encourage the LLM to “hallucinate” and provide an answer to the question
    - Embed the hallucinated answer
    - Find content that is semantically similar
    - HyDE improves recall but can reinforce incorrect assumptions if the hallucinated answer is wrong
- Multi-Query Retrieval
    - Generating 3–5 variations of the user's question to capture different parts of the vector space.

When RAG should say “I don’t know”

- Silence is better than hallucination
- Implement:
    - Confidence thresholds
    - Empty retrieval handling
    - Abstention as a feature, not a failure
- Advanced systems optimize for trust, not verbosity

Human-in-the-Loop in RAG

Why HITL exists

- Retrieval can be wrong but confident
- Generation amplifies retrieval errors
- Fully automated RAG optimizes fluency, not trust
- **Core idea**
    - Use humans only where uncertainty is highest
    - Treat humans as a control mechanism, not a crutch
    - Human feedback becomes training and evaluation data over time

Where humans intervene

- After retrieval. Validate relevance
- Before generation. Prune or reorder context
- After generation. Approve, correct, or escalate
- **Key pattern**

    - Low confidence → abstain → human review.
    - “I don’t know” is a feature.

Vector Databases in RAG

What a vector DB actually does

- Stores embeddings
- Runs approximate nearest neighbor search
- Filters using metadata
- Trades recall for speed
- **Important**
    - The database does not fix bad embeddings
    - RAG quality is decided upstream
    - Index configuration and metadata schema matter more than ANN algorithm choice

Putting It Together

High-quality RAG systems

- Use asymmetric embeddings
- Combine vectors with metadata
- Apply confidence thresholds
- Escalate to humans when needed
- **Mental model**
    - RAG is a system with feedback loops.
    - Not a prompt and not just a vector search.

Resources

https://docs.cloud.google.com/vertex-ai/generative-ai/docs/embeddings/task-types
https://arxiv.org/abs/2004.04906
https://arxiv.org/abs/2212.10496
https://www.pinecone.io/learn/retrieval-augmented-generation/
https://www.microsoft.com/en-us/research/publication/guidelines-for-human-ai-interaction/